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Portfolio hedging and basis risks

KIAN-GUAN LIM

Department of Finance and Banking, Faculty of Business Administration, National University of Singapore, 10 Kent Ridge Crescent, Singapore 0511

Minimum variance hedged portfolios using futures are formed by taking the linear projection of spot price changes onto futures price movements as the hedge ratio. This unwittingly assumes that the underlying spot–futures price movements follow a cointegrated process, given that the spot and the futures prices are integrated processes. If the spot–futures prices are not cointegrated, the hedged portfolio suffers from the risk of potentially large changes in its value. Empirical findings using the Nikkei stock index and the Nikkei 225 futures show this deviation in intraday trading prices. The basis movements which have often been used by intraday traders to predict future price changes, are tested to be mostly unit root processes. This is shown to be due largely to non-cointegration of the spot–futures prices, and suggests why it is profitable to trade futures using basis knowledge only if trading is done on a continual basis.

I. INTRODUCTION

Hedgers of price risks in financial assets have to determine the amount of futures contract to buy or sell to insure against equity portfolio risks. Obviously the optimal amount depends on the hedger's risk preference (Stulz, 1984; Figlewski, 1985). More commonly, the minimum variance hedge ratio is assumed (Ederington, 1979; and Frankle, 1980), and such hedged portfolios are then examined to see if *ex post* risk is minimal. In determining the minimum variance hedge ratio, spot price changes are linearly projected onto the futures price changes, and the least squares estimate of the slope coefficient is used.

I show that, if the spot–futures prices are not cointegrated, the hedged portfolio suffers from the risk of potentially large changes in its value, which is not well recognized. Existing studies limit themselves to basis risks; see a thorough study in Figlewski (1984). Basis risk appears to have been explicitly or implicitly modelled as a discrete stationary process. This paper also shows how it is possible that the basis could also be an integrated process and relates it to the cointegration econometrics framework developed below. Empirical testing is performed and implications are then drawn about hedged portfolio value and basis trading regimes.

I shall firstly set up the usual nomenclature for the econometric modelling of spot–futures relationship. Let S denote the spot price level, F the futures price level, and assume

a general linear process

$$S_t = \alpha + \gamma t + \beta F_t + \varepsilon_t$$

where α , γ , and β are constants, t is a time index, ε is a mean-zero residual error and the subscript t indicates realization in period t . The first differences in the levels are related as

$$\Delta S_t = \gamma + \beta \Delta F_t + \Delta \varepsilon_t$$

The value change in the minimum variance hedge portfolio can then be represented by $\Delta S_t - \beta \Delta F_t$, or $\gamma + \Delta \varepsilon_t$ in each period t .

Over an interval of time, say $n + 1$ periods, the cumulative value change is

$$(n + 1)\gamma + \sum_{t=j}^{j+n} \Delta \varepsilon_t$$

If $\Delta \varepsilon_t$ is stationary and independently distributed, the variance of the cumulative value change increases linearly with time. The implication is that the hedged portfolio value follows a random walk. If $\Delta \varepsilon_t$ is stationary but serially correlated, it is possible for the variance of the hedged portfolio value change to exhibit non-linear patterns over time.

Finance literature has gathered much empirical evidence about price levels being unit root processes (Baillie and Bollerslev, 1989; Corbae and Ouliaris, 1988; Corbae, Lim

and Ouliaris, 1992). Suppose S_t and F_t are unit root or $I(1)$ processes, then an interesting issue is whether or not S_t and F_t are cointegrated with cointegrating vector $(1, -\beta)$. If S_t and F_t are not cointegrated, then ε_t is typically an $I(1)$ process and $\Delta\varepsilon_t$ is stationary though not necessarily independently distributed. This leads to the hedged portfolio value following a random walk.

However, if S_t and F_t are cointegrated, ε_t is itself stationary though not necessarily independent over time. The hedged portfolio's cumulative value change can then be characterized as

$$(n+1)\gamma + \sum_{t=j}^{j+n} (\varepsilon_{t+1} - \varepsilon_t)$$

or $(n+1)\gamma + \varepsilon_{j+n+1} - \varepsilon_j$. Whatever the length of time, i.e. whatever n may be, the variance or risk of this cumulative change is constant. Thus the hedged portfolio value will not deviate with increasing unpredictability over time.

The above arguments show clearly the necessity of performing a cointegrating regression analysis of the spot-futures price levels. This will help determine whether hedged portfolios are of any use in preserving capital value over any interval of time. In the following section, I discuss the cointegration theory and motivate the use of the canonical cointegrating regression (CCR) technique developed by Park (1988). Section III reports tests of the unit root hypotheses of the spot and futures price levels and the basis for the Nikkei 225 index and the futures contract. Section IV contains results of the CCR and also shows a robust result concerning the $I(1)$ nature of the basis. Apart from implications on portfolio hedging, the results also have implications for intraday trading and suggest how past basis movements could be exploited to yield profitable information on the future basis change. Section V contains the conclusions.

II. COINTEGRATING REGRESSION

This section contains a brief review of cointegration theory (Engle and Granger, 1987) and relates the theory to the regression methodology employed in this paper.

A cointegrated regression may be written as

$$S_t = \omega_1(\theta_1, t) + \beta'F_t + u_t$$

where $\omega_1(\theta_1, t)$ is a deterministic function assumed to depend on time t and is parametrized by θ_1 . β is an m -vector of constants and the m -vector process F_t evolves according to

$$F_t = \omega_2(\theta_2, t) + F_{t-1} + v_t$$

where $\omega_2(\theta_2, t)$ is an m -vector deterministic function. Moreover, u_t and v_t are random variables of stationary processes. Therefore, $(S_t, F_t)'$ is an $(m+1)$ -vector integrated order one, $I(1)$, process. In the spot-futures relationship that we are examining, $m = 1$.

The usual alternative approach to regression involving these levels is to apply OLS regression involving first differences. However, the cointegrating regression technique yields significant advantages over the conventional econometric methodology such as OLS on first differences.

Stock (1987) and Park and Phillips (1988) have shown that, when OLS is applied to the cointegrated regression, the parameter estimates converge to the true parameter β at the rate T (sample size), faster than $\sqrt{T}$ for the conventional case. Faster convergence occurs because the sample moments of the component F_t are an order of magnitude larger than the sample moments of u_t , and this dominates the impact of any correlation that may exist between the innovations in F_t and the true error term u_t . This so-called superconsistency result continues to hold even in the presence of contemporaneous correlation in u_t , endogeneity of F_t and stationary measurement error in S_t and F_t .

However, the advantages in favour of OLS disappear if S_t and F_t do not form a cointegrated system. Specifically, when u_t is itself an integrated process, i.e. S_t and F_t are not cointegrated, then applying OLS leads to 'spurious' regression results (Granger and Newbold, 1974; Phillips, 1986). In particular, OLS parameter estimates, though unbiased, are inconsistent when u_t is an integrated process, and the conventional t - and F -ratio statistics do not possess limiting distributions. Also, the Durbin-Watson statistic converges in probability to zero, whereas R^2 has a non-degenerate limiting distribution as the sample size approaches infinity.

I now present an applied methodology, as in Corbae *et al.* (1992), for the estimation and testing of cointegrated regressions. Since the existing literature maintains priors of $I(1)$ about the price processes and maintains that spot and futures prices are cointegrated, I employ test statistics that use these priors as null hypotheses. This has the effect of minimal type I error. It also explains why some other cointegration tests that are based on different nulls are not employed here.

Step 1 Pretest the time-series data S_t and F_t to determine if the data are consistent with the $I(1)$ hypothesis. More generally, a time trend is allowed in the $I(1)$ specifications. Park and Choi (1988) suggested a J -statistic for the unit root test which was shown to have minimum size distortion bias. The Z_α and the Z_t statistics (Phillips, 1987) are also used for comparison. An application of the J -statistic is reported in Lim and Phoon (1991).

The J -statistic is defined as

$$J(p, q) = (RSS_p - RSS_q) / RSS_q \quad \text{for } 0 < p < q$$

where RSS_k for $k = p, q$ is the residual sum of squares from the regression of S_t on a constant, S_{t-1} , and time variables $t, t^2, t^3, \dots, t^k$. Intuitively, this statistic is a test of the restrictions of the $(p+1)$ th to q th terms of the time polynomial being zero. If fitting higher time polynomials is able

to explain variations in S_t , the resulting higher value of the J -statistic indicates the existence of a unit root.

Step 2 Estimate the coefficient β using the canonical cointegrating regression (CCR) technique developed by Park (1988). Then test whether S_t and F_t are cointegrated using Park's (1988) $H(p, q)$, $q > p$, statistic. This statistic is based on a null hypothesis of cointegration, i.e. stationary error, allowing for deterministic time trend up to the p th order. If the regression is cointegrated with irrelevant polynomial time trends, the application of CCR should yield insignificant coefficients on these variables. The $H(p, q)$ statistic also possesses an asymptotic χ^2_{q-p} distribution.

The CCR regression technique is based on a GLS-type correction to the original data; Park (1988) gives the details of the technique. The transformation is designed to yield an estimate for β that permits conventional hypothesis testing. In particular, hypothesis testing may be conducted using conventional asymptotic χ^2 statistics. Park (1988) develops the CCR procedure under mild regularity conditions for the innovation sequences driving S_t and F_t .

Moreover, the CCR estimator has the advantage that the error process is specified in a non-parametric fashion, and can easily accommodate deterministic non-stationarity in S_t and F_t such as that arising from a constant term and a time trend.

III. PRICE PROPERTIES OF THE NIKKEI INDEX AND FUTURES

Research on the Japanese stock market and the Nikkei 225 futures market has been gaining momentum over the last several years (Bailey, 1989; Brenner *et al.*, 1989; Lim, 1992a; Ziemba, 1990). With the wide price swings in the Japanese stock market, whereby in intraday trades it is common for the Nikkei 225 index to swing several hundred points either way, or up to 5% of the total index value, hedging using the Nikkei 225 stock index futures is a significant activity. The futures contract was first traded in the Singapore International Monetary Exchange in 1986 and also in the Osaka Exchange in 1989.

Using intraday futures trading data over 20 days at 5-minute intervals and matching it with the Nikkei 225 index reported at these same intervals, unit root tests were performed on the time series properties of the index level, the futures price level and the basis. The 20 trading days were randomly selected from four contracts: June 1988, September 1988, June 1989 and September 1989. Each daily data set comprises time-stamped transaction records showing the traded prices. At each 5-minute interval throughout the trading day, the traded price just before the end of the interval was matched with the 'QUICK'-reported Nikkei spot index at the end of the interval.

The computed test statistics are based on the null hypotheses of unit roots for the price levels and the basis; they are the J -statistic developed by Park and Choi (1988), and the Z_α and Z_t statistics of Phillips (1987). The empirical results are reported in Table 1. Statistics that fall below the critical value indicate rejection of the null at that significance level.

Table 1 shows that the futures prices behave as $I(1)$ process except for day 17 when both the Z_α and Z_t statistics indicate rejection of the null at 1% significance. The spot prices also behave as an $I(1)$ process except for days 3 and 12, when the Z_t statistics indicate rejection of the null at 1% significance. The basis appears to follow an $I(1)$ process for the most part, although it could well be stationary on some days, e.g. days 4, 9, and 12 when the Z_t statistics indicate rejection of the null at 1% significance. It would appear that the Z_t statistic is more powerful than the Z_α and J -statistics, though Lim and Phoon (1991) discussed the bias towards rejection in the Z -statistics relative to the J -statistics.

IV. COINTEGRATION TESTS

In this section, I use the canonical cointegration regression of Park (1988). The following regression is performed:

$$S_t = \alpha + \gamma t + \beta F_t + \varepsilon_t$$

The CCR technique yields consistent estimates of α , γ and β provided that ε_t is stationary. Consistency still obtains even if F_t and ε_t are not independent, as this bias reduces to zero asymptotically. Such desirable features of estimation using CCR cannot be obtained with OLS regression on the first differences. As indicated in the previous section, that S_t and F_t are typically $I(1)$ processes provides the justification for applying the CCR technique.

The intraday spot prices are regressed against a constant, time, and the matched futures prices for each of the 20 trading days. The $H(p, q)$ statistic developed by Park (1988) to test for deviation from the null of cointegration is computed in each regression. Based on the null of a linear time trend and fitting additional time polynomials to degree 3 yields a χ^2 distribution with two degrees of freedom for the H -statistic under the null. The estimates and test statistics are reported in Table 2.

From the CCR of spot prices on futures prices, the H -statistics indicate rejection of the null of cointegration for days 12, 15, 18 and 20 at 1% significance, and for days 1, 3, 6 and 9 at 5% significance. In other words, there are many days (40% of my sample) in which a hedged portfolio value behaves like a random walk. In these cases, the portfolios are unlikely to offer protection against extreme movements in spot prices.

On the other days, when the null of cointegration is not rejected, the minimum variance hedge ratios are estimated to range from about 0.2 to 1.4 with the majority of cases

Table 1. Unit root tests of futures prices, spot prices, and basis of the Nikkei 225 contract using the J-statistic and Phillips' Z-statistics

Contract	$J(1, 5)$	$Z_{\alpha}(1, 5)$	$Z_t(1, 5)$	Contract	$J(1, 5)$	$Z_{\alpha}(1, 5)$	$Z_t(1, 5)$
June 1988				June 1989			
Trading day 1				Trading day 11			
Futures	0.956	− 19.065	− 3.788 ^a	Futures	0.179	− 19.414	− 3.432
Spot	6.509	− 7.752	− 2.058	Spot	2.593	− 20.314	− 3.344
Basis	8.163	− 2.720	− 0.683	Basis	2.994	− 19.235	− 3.131
Trading day 2				Trading day 12			
Futures	2.926	− 12.000	− 1.996	Futures	0.977	− 14.601	− 2.715
Spot	0.609	− 14.320	− 2.946	Spot	2.401	− 23.772	− 4.948 ^b
Basis	0.604	− 14.043	− 2.760	Basis	4.845	− 18.902	− 4.707 ^b
Trading day 3				Trading day 13			
Futures	0.254 ^a	− 21.490 ^a	− 3.740 ^a	Futures	4.438	− 5.226	− 1.795
Spot	2.149	− 25.163 ^a	− 4.845 ^b	Spot	5.624	− 4.826	− 1.597
Basis	3.728	− 15.451	− 3.438	Basis	2.039	− 17.081	− 3.604 ^a
Trading day 4				Trading day 14			
Futures	1.219	− 8.613	− 2.091	Futures	1.361	− 13.803	− 2.965
Spot	3.141	− 15.621	− 4.003 ^a	Spot	6.388	− 15.279	− 3.592 ^a
Basis	8.910	− 18.616	− 5.389 ^b	Basis	4.398	− 17.055	− 3.866 ^a
Trading day 5				Trading day 15			
Futures	6.485	− 2.061	− 0.720	Futures	3.252	− 8.862	− 2.393
Spot	9.761	− 2.580	− 0.994	Spot	3.153	− 14.781	− 3.993 ^a
Basis	2.596	− 5.275	− 1.312	Basis	2.361	− 16.615	− 3.607 ^a
September 1988				September 1989			
Trading day 6				Trading day 16			
Futures	1.267	− 10.830	− 2.823	Futures	0.630	− 17.225	− 3.581 ^a
Spot	1.470	− 6.940	− 1.464	Spot	1.498	− 11.049	− 2.497
Basis	3.924	− 14.183	− 3.115	Basis	2.762	− 9.055	− 2.426
Trading day 7				Trading day 17			
Futures	0.445	− 21.794 ^a	− 3.758 ^a	Futures	0.336	− 30.956 ^b	− 4.895 ^b
Spot	4.766	− 6.428	− 2.210	Spot	0.157 ^a	− 22.538 ^a	− 3.828 ^a
Basis	1.291	− 13.301	− 3.229	Basis	0.148	− 21.302 ^a	− 3.926 ^a
Trading day 8				Trading day 18			
Futures	2.240	− 7.386	− 1.443	Futures	24.598	− 4.101	− 1.487
Spot	1.864	− 17.008	− 3.882 ^a	Spot	1.363	− 10.051	− 2.569
Basis	2.121	− 19.062	− 3.677 ^a	Basis	1.209	− 11.377	− 2.721
Trading day 9				Trading day 19			
Futures	1.646	− 6.314	− 1.797	Futures	1.802	− 11.630	− 2.305
Spot	3.373	− 8.718	− 2.996	Spot	1.439	− 15.922	− 3.568 ^a
Basis	1.222	− 20.087	− 4.808 ^b	Basis	2.376	− 13.831	− 2.460
Trading day 10				Trading day 20			
Futures	1.642	− 14.631	− 3.379	Futures	5.323	− 4.411	− 1.274
Spot	1.584	− 11.113	− 2.482	Spot	1.530	− 8.342	− 1.821
Basis	2.024	− 19.730	− 4.017 ^a	Basis	0.369	− 12.938	− 2.891

Notes: Critical values 1%, 5%
J-statistic 0.117, 0.294
 Z_{α} statistic − 27.702, − 20.724
 Z_t statistic − 4.144, − 3.540

^a For $H_0: I(1)$; reject null at 5% significance level.
^b For $H_0: I(1)$ reject null at 1% significance level.

being close to but below 1. To confirm this finding, an OLS regression of spot price changes on futures price changes was also performed. The results are reported in Table 3. The t -statistics of the slope coefficient estimators show clearly that spot price changes and futures price changes are significantly positively correlated. The coefficient estimates are similar in magnitudes to those in the CCR. This con-

firms the CCR findings. The significant positive correlations between stock index price changes and the corresponding futures price changes are also reported in other studies such as Kawaller *et al.* (1987) and Lim (1991b). Using the CCR estimates reported in Table 2, I also test if these estimates are significantly different from unity. The test statistics and p -values are reported in Table 4. The null

Table 2. Canonical cointegrating regression of spot price on futures price of the Nikkei 225 contract

Contract	Constant	Time	Futures	χ^2	Contract	Constant	Time	Futures	χ^2
June 1988					June 1989				
Trading day 1					Trading day 11				
Coefficients	1406.336	495.935	0.944	9.217	Coefficients	26557.046	− 92.075	0.172	2.471
p-value	(0.452)	(0.026)	(0.021)	(0.010) ^a	p-value	(0.022)	(0.665)	(0.331)	(0.291)
Trading day 2					Trading day 12				
Coefficients	− 711.757	157.332	1.025	2.390	Coefficients	2731.782	− 857.874	0.899	9.992
p-value	(0.476)	(0.131)	(0.013)	(0.303)	p-value	(0.392)	(0.001)	(0.002)	(0.007) ^b
Trading day 3					Trading day 13				
Coefficients	7169.333	321.770	0.728	6.733	Coefficients	− 14130.316	353.290	1.419	2.839
p-value	(0.072)	(0.015)	(0.000)	(0.035) ^a	p-value	(0.002)	(0.063)	(0.000)	(0.242)
Trading day 4					Trading day 14				
Coefficients	9106.653	311.118	0.658	5.764	Coefficients	1359.120	274.749	0.952	4.397
p-value	(0.168)	(0.067)	(0.035)	(0.056)	p-value	(0.448)	(0.074)	(0.002)	(0.111)
Trading day 5					Trading day 15				
Coefficients	10936.946	− 165.030	0.592	3.245	Coefficients	23446.772	842.053	0.295	10.597
p-value	(0.000)	(0.021)	(0.000)	(0.197)	p-value	(0.001)	(0.004)	(0.079)	(0.005) ^b
September 1988					September 1989				
Trading day 6					Trading day 16				
Coefficients	12502.338	236.758	0.558	8.680	Coefficients	9978.626	− 642.750	0.711	3.955
p-value	(0.031)	(0.063)	(0.010)	(0.013) ^a	p-value	(0.388)	(0.051)	(0.242)	(0.138)
Trading day 7					Trading day 17				
Coefficients	3846.225	− 289.014	0.868	3.074	Coefficients	− 1700.521	130.498	1.041	3.865
p-value	(0.401)	(0.127)	(0.060)	(0.215)	p-value	(0.386)	(0.119)	(0.000)	(0.145)
Trading day 8					Trading day 18				
Coefficients	7455.679	95.913	0.733	3.320	Coefficients	696.050	− 261.977	0.973	34.272
p-value	(0.254)	(0.349)	(0.037)	(0.190)	p-value	(0.394)	(0.000)	(0.000)	(0.000) ^b
Trading day 9					Trading day 19				
Coefficients	3916.615	− 1151.798	0.860	6.703	Coefficients	24474.287	− 107.415	0.290	2.686
p-value	(0.233)	(0.010)	(0.000)	(0.035) ^a	p-value	(0.000)	(0.204)	(0.047)	(0.261)
Trading day 10					Trading day 20				
Coefficients	8154.007	130.715	0.699	1.420	Coefficients	24290.606	− 168.540	0.289	11.425
p-value	(0.173)	(0.269)	(0.015)	(0.492)	p-value	(0.000)	(0.069)	(0.004)	(0.003) ^b

Notes: H_0 : cointegration; test statistic is $H(1, 3) = \chi^2$ d.f. 2; reject H_0 given one-tailed test if H is too large.

^a Reject H_0 at 5% significance level.

^b Reject H_0 at 1% significance level.

hypothesis of unit slope cannot be rejected at the 5% significance level for 13 out of the 20 days. If I eliminate the cases of non-cointegration, as discussed earlier, the null cannot be rejected in 8 out of 12 days. This turns out to be about 67% of the cases. This means that both price changes are highly correlated.

This result has certain implications on the basis as defined by the difference of futures price and the spot price. The basis is then

$$(1 - \beta)F_t - \alpha - \gamma t - \varepsilon_t$$

If β is unity, the basis is stationary except with a deterministic trend provided that the spot–futures prices are cointegrated. This happened in 8 out of 20 days using our sample. If β is not unity in the cointegrated regression, the basis

possesses a unit root as F_t is an $I(1)$ process. The results in Table 4 show that β is often close, if not equal to, unity. Thus, any significant appearance of a unit root in the basis is due more to the fact that the spot and futures prices are not cointegrated. This suggestion is corroborated by the empirical evidence presented in Table 1 that the basis is mostly an $I(1)$ process.

If the regression is not cointegrated, then whether or not β is unity, the basis possesses a unit root due to ε_t being $I(1)$. In trading days 1, 3, 6, 18 and 20, in which non-cointegration is detected as in Table 2, the basis is also shown to be $I(1)$ as reported in Table 1. And in trading days 5, 6, 11, 19 and 20, in which β is unlikely to be unity, as reported in Table 4, the bases as reported in Table 1 are also shown to be $I(1)$. Empirically, these two possibilities of either

Table 3. Estimating optimal futures hedging coefficients of the Nikkei 225

Contract	Constant	β	F-value	Contract	Constant	β	F-value
June 1988				June 1989			
Trading day 1				Trading day 11			
Coefficients	0.935	0.910	15.482	Coefficients	- 1.788	0.244	1.073
p-value	(0.296)	(0.000)	(0.000)	p-value	(0.278)	(0.106)	(0.351)
Trading day 2				Trading day 12			
Coefficients	- 0.897	0.898	11.196	Coefficients	- 6.158	0.869	12.450
p-value	(0.316)	(0.000)	(0.000)	p-value	(0.057)	(0.000)	(0.000)
Trading day 3				Trading day 13			
Coefficients	2.296	0.741	26.099	Coefficients	2.022	0.477	6.707
p-value	(0.084)	(0.000)	(0.000)	p-value	(0.271)	(0.000)	(0.003)
Trading day 4				Trading day 14			
Coefficients	- 0.109	0.329	2.356	Coefficients	4.105	0.261	2.800
p-value	(0.528)	(0.024)	(0.107)	p-value	(0.087)	(0.066)	(0.071)
Trading day 5				Trading day 15			
Coefficients	- 0.775	0.305	16.787	Coefficients	- 0.370	0.502	8.473
p-value	(0.735)	(0.000)	(0.000)	p-value	(0.438)	(0.000)	(0.001)
September 1988				September 1989			
Trading day 6				Trading day 16			
Coefficients	0.466	0.414	13.898	Coefficients	- 3.171	0.483	1.168
p-value	(0.341)	(0.000)	(0.000)	p-value	(0.242)	(0.081)	(0.325)
Trading day 7				Trading day 17			
Coefficients	- 1.308	0.443	14.119	Coefficients	- 1.265	0.957	14.145
p-value	(0.847)	(0.000)	(0.000)	p-value	(0.397)	(0.000)	(0.000)
Trading day 8				Trading day 18			
Coefficients	- 4.854	0.512	22.159	Coefficients	- 4.634	0.827	2.499
p-value	(0.008)	(0.000)	(0.000)	p-value	(0.280)	(0.019)	(0.105)
Trading day 9				Trading day 19			
Coefficients	- 4.367	0.335	4.861	Coefficients	- 5.292	0.287	5.923
p-value	(0.132)	(0.003)	(0.012)	p-value	(0.014)	(0.010)	(0.006)
Trading day 10				Trading day 20			
Coefficients	2.870	0.548	6.628	Coefficients	- 1.758	0.381	1.162
p-value	(0.190)	(0.000)	(0.003)	p-value	(0.385)	(0.075)	(0.331)

non-cointegration or else non-unit β explain most of the cases of $I(1)$ basis.

When the basis b_t follows a unit root process, I can write

$$b_t = b_{t-1} + \eta_t$$

where η_t is a residual error not correlated with b_{t-1} , but is not necessarily mean zero or independent over time. This implies that intraday traders can obtain the best prediction about future basis movement by taking note of the past basis movements. Information on the future basis changes are fully incorporated in the current basis. However, it is imperative that such traders watch and trade on a continual basis in order to profit from the past basis information. This is because, if the interval for forecast of future basis is widened, then since b_t is an $I(1)$ process, its variance increases over time. This increased variance introduces unnecessary uncertainties and risk for traders. The risk is minimized when the forecast is made on a continual basis. That

this indeed is the case explains the proliferation of individual trades (also called independents on the SIMEX exchange floor). These trades are typically in small orders and the traders would buy and sell continually. Of course, if b_t is stationary, then whatever the interval of forecast, the uncertainty remains the same.

V. CONCLUSION

I have raised an issue regarding the optimal hedging of spot positions using futures. Employing the Nikkei 225 stocks and the Nikkei 225 futures, there is evidence that the spot-futures prices are not necessarily cointegrated on some trading days even though their first differences are stationary. Non-cointegration in the price levels implies that the value of the hedged portfolio, in particular the minimum variance hedged portfolio, follows an $I(1)$ process, or more

Table 4. Canonical cointegrating regression of spot price on futures price: a test of unit optimal hedge ratio of the Nikkei 225 contracts

Trading day	t-statistic	p-value
1	- 0.124	(0.451)
2	0.057	(0.477)
3	- 1.470	(0.074)
4	- 0.961	(0.171)
5	- 6.490	(0.000) ^b
6	- 1.892	(0.032) ^a
7	- 0.242	(0.405)
8	- 0.664	(0.255)
9	- 0.707	(0.242)
10	- 0.974	(0.168)
11	- 2.114	(0.020) ^a
12	- 0.329	(0.372)
13	2.936	(0.003) ^b
14	- 0.155	(0.439)
15	- 3.432	(0.001) ^b
16	- 0.288	(0.388)
17	0.250	(0.402)
18	- 0.364	(0.360)
19	- 4.215	(0.000) ^b
20	- 7.234	(0.000) ^b

Note: H_0 : slope coefficient = 1.
^a Reject H_0 at 5% significance level.
^b Reject H_0 at 1% significance level.

specifically, a random walk. Such a price risk has not been well known in the existing literature on hedging, as far as I know.

I also provide unit root tests showing evidence that the basis is mostly an $I(1)$ process. This is consistent with both cases of spot-futures cointegration or non-cointegration, provided the slope coefficient of the spot price on futures price is not unity. Even if the slope is unity, non-cointegration in the spot-futures prices would induce an $I(1)$ basis. A basis that possesses a random walk, a special case of $I(1)$, implies that it is most profitable for intratraders to use basis knowledge to trade on a continual basis.

This study contributes to the existing literature by characterizing hedged portfolio as well as basis behaviours in the cointegration framework. Empirical tests on the cointegration framework are performed and interesting financial implications are drawn.

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